

View Reviews

Paper ID

637

Paper Title

SPASHTA — Speech Posthoc Attribution with Salient Highlighting for Transparent Analysis

Track Name

CVIP2025-Round2

Reviewer #1

Questions

1. Is the paper relevant to the scope of the conference (Computer Vision, Image Processing, or related)?

Yes

2. Does the paper follow the prescribed Springer format/template?

Yes

3. Briefly summarize the paper in your own words. You may write : what problem is solved, and what methods are used by authors to solve the problems/challenges. What are their key contributions. ((Avoid using LLM for this and other questions. Also avoid copying from the abstract — this helps verify reviewer understanding.)

This paper presents new method SPASHTA related to decision-making patterns of Speech Emotion Recognition (SER) models. It is relevant to current research. seems a novel idea and may be accepted

4. Comment on how proposed approach is validated, what dataset is used, soundness of validation. (Comment briefly if experimental results and analysis are sufficient and convincing?)

through experiments. method seems good

5. What are the strengths of the paper?

novelty

6. Comment on weakness and/or limitations of the paper and you may provide suggestions for authors for improvement

must contain the flow chart to understand the flow of data

7. Final Recommendation

Weak Accept

Questions

1. Is the paper relevant to the scope of the conference (Computer Vision, Image Processing, or related)?

Yes

2. Does the paper follow the prescribed Springer format/template?

Yes

3. Briefly summarize the paper in your own words. You may write : what problem is solved, and what methods are used by authors to solve the problems/challenges. What are their key contributions. ((Avoid using LLM for this and other questions. Also avoid copying from the abstract — this helps verify reviewer understanding.)

This work introduces a model-agnostic framework that generates a listenable explanation by masking the non-salient audio features. The proposed method leverages MFCC features to assess and evaluate the model behavior, revealing that it is inherently suitable for low-resource settings such as spoken Indic languages, as the model uses prosodic cues rather than semantics. At last, the paper discusses the interpretability of the explanations validated by human subject experts.

4. Comment on how proposed approach is validated, what dataset is used, soundness of validation. (Comment briefly if experimental results and analysis are sufficient and convincing?)

The proposed method is validated on a small in-house dataset (BhavanaVani) and a Hindi SER dataset from Kaggle. While the in-house dataset is limited and biased, raising concerns about generalizability, the method is evaluated empirically using quantitative metrics and human subject evaluations. The single model architecture and absence of baseline comparisons make the validation less comprehensive and partially convincing.

5. What are the strengths of the paper?

1. The manuscript is well-structured and organized, highlighting the literature gap, novelty, and contributions.
2. The proposed method provides an intuitive way for humans to understand model behavior by generating listenable audio explanations, making it a significant and practical contribution to the field of XAI for speech processing.
3. The proposed method is model-agnostic and applicable to SER models without requiring architectural changes.
4. Curation of the BhavanaVani dataset and validation of the proposed method using a low-resource Indic language address practical challenges.
5. The authors' approach to validate interpretability and meaningfulness to humans empirically is a core strength of the paper.

6. Comment on weakness and/or limitations of the paper and you may provide suggestions for authors for improvement

1. The paper does not discuss the choice of perturbation parameters. Authors are suggested to justify the window size or stride chosen and how they arrived at it.
2. The in-house dataset has some known biases, which raise concerns about the generalizability; the authors are suggested to discuss the methods to mitigate those biases.
3. There is no mention of the computational overhead of the perturbation-based approach. For real-time applications, this could be a significant factor, and a discussion on the efficiency of the method would be beneficial.
4. The method considers only a single CNN-GRU-Attention model architecture, yet claims to be model-agnostic. Lack of model comparisons lowers the confidence in the findings. The inclusion of quantitative comparison with different model architectures, as well as baseline explainability methods like LIME or SHAP, would strengthen the claims and highlight the identified literature gap and contributions.

7. Final Recommendation

Accept

Reviewer #4

Questions

1. Is the paper relevant to the scope of the conference (Computer Vision, Image Processing, or related)?

Partially

2. Does the paper follow the prescribed Springer format/template?

Yes

3. Briefly summarize the paper in your own words. You may write : what problem is solved, and what methods are used by authors to solve the problems/challenges. What are their key contributions. ((Avoid using LLM for this and other questions. Also avoid copying from the abstract – this helps verify reviewer understanding.)

The authors address the interpretability of Speech Emotion Recognition (SER) low resource and real-world scenarios. In this work they employed a posthoc explainability technique that can recognize low-resource model with the help of interpretability metrics. They take into consideration various factor such as speaker speech style, region, and other. Furthermore, the authors have overcome some of the issues of the literature using masking-based methodology.

4. Comment on how proposed approach is validated, what dataset is used, soundness of validation. (Comment briefly if experimental results and analysis are sufficient and convincing?)

The authors claims that they employed interpretability metrics such as average gain/drop metric are briefed, but the result of these metrics are not included. Though precision, recall and F-Score are used as metrices, but seems inadequate to support the prosed method efficiency.

5. What are the strengths of the paper?

The strength are very limited, some of them are:

1. The solution provided is for low-resource model, however, the claim is not connivingly justified by experimental results.
2. They take into consideration various factor of speakers span over speech style, region, and other.

6. Comment on weakness and/or limitations of the paper and you may provide suggestions for authors for improvement

The major weakness are:

1. In table 1, Region-wise Distribution of Samples in BhavanaVani Dataset is illustrate. However, the sample size of the data is just 47 and total duration is mere 22 minutes. In this case they have used Speech Emotion Recognition Hindi (Kaggle Dataset) to train the model. However the technical details of overfitting and underfitting are not included. Also, the split of training and test data are explained.
2. For evaluations hybrid CNN-GRU-Attention model is employed, the authors are required to either cite the work or provide the architecture design of the model in section 3.
3. In Figure 3, the y-axis (ordinate) title. If could add that, it will technically make it meaningful.
4. The work is having limited mathematical explanation and technically weak. Also, the results are not very conclusive.

7. Final Recommendation

Reject